

BUSINESS REQUIREMENT DOCUMENT (BRD)

E-Commerce Order Management Database System

Business Objectives

The main objective of the E-Commerce Order Management Database System is to develop a centralized database that efficiently manages customer information, product details, supplier records, orders, payments, shipments, and customer reviews. The system aims to improve business efficiency, reduce manual work, minimize data redundancy, and ensure accurate information for decision-making.

Business Requirements

The system must fulfill the following business requirements:

- Maintain customer registration and profile information.
- Store and manage product details.
- Organize products into categories.
- Maintain supplier information.
- Process customer orders accurately.
- Record payment details.
- Track shipment status.
- Store customer reviews and ratings.
- Generate reports for business management.
- Maintain data consistency and integrity.

Stakeholder Roles

Customer – Registers, places orders, makes payments, tracks shipments, and gives reviews.

Administrator – Manages customers, products, categories, suppliers, orders, payments, and reports.

Supplier – Supplies products and updates stock availability.

Delivery Staff – Delivers orders and updates delivery status.

Payment Gateway – Processes online payments securely.

Business Rules

The following business rules apply:

- Every customer must have a unique Customer ID.
- Each product belongs to one category.
- A customer can place multiple orders.
- An order must contain at least one product.
- Every payment must be linked to one order.
- Every shipment must correspond to one order.
- Customers can review only purchased products.
- Stock quantity must be updated after order confirmation.

Business Process

The business process follows these steps:

1. Customer Registration
2. Product Browsing
3. Product Selection
4. Order Placement
5. Payment Processing
6. Order Confirmation
7. Shipment Processing
8. Delivery
9. Customer Review
10. Report Generation

Scope

The project includes:

- Customer Management
- Product Management
- Category Management
- Supplier Management
- Order Management
- Payment Management
- Shipment Management
- Review Management
- Business Report Generation

The project does not include marketing, accounting, or employee payroll management.

Risks

Potential risks include:

- Data loss due to system failure.
- Unauthorized access to customer data.
- Payment transaction failures.
- Network connectivity issues.
- Incorrect inventory updates.
- Delay in shipment tracking updates.

Assumptions

- Customers have internet access.
- Payment gateway services are available.
- Product information is updated regularly.
- Suppliers provide accurate stock information.
- Users enter valid data.

Constraints

- Only the predefined core entities are used.
- The system depends on internet connectivity.
- Database access is restricted to authorized users.
- The project follows the semester guidelines.
- Data security policies must be followed.

Conclusion

The Business Requirement Document defines the business needs and expectations of the E-Commerce Order Management Database System. It identifies the objectives, stakeholders, business rules, processes, risks, assumptions, and constraints. This document serves as the foundation for designing and developing a reliable database system that supports efficient business operations.